

Python & Data Science Practical Lab Question Paper

Section A – Python Basics (Variables, Data Types, Operators)

1. Write a program to swap two variables without using a third variable.
2. Create variables of different data types and print their types.
3. Perform arithmetic operations on two user-input numbers.
4. Write a program to calculate simple interest.
5. Convert temperature from Celsius to Fahrenheit.
6. Demonstrate type casting with user input.
7. Check whether a number is positive, negative, or zero.
8. Find the largest of three numbers using conditional statements.

Section B – Conditional & Looping Statements

9. Print numbers from 1 to 50 using a for loop.
10. Print the multiplication table of a given number.
11. Find the sum of first N natural numbers.
12. Check whether a number is prime.
13. Print Fibonacci series up to N terms.
14. Reverse a given number using a while loop.
15. Count vowels in a given string.
16. Print a right-angled star pattern.

Section C – File Handling

17. Create a text file and write 5 lines into it.
18. Read contents of a file and display them.
19. Count number of words in a file.
20. Count number of lines in a file.
21. Append data to an existing file.
22. Copy contents from one file to another.

23. Search for a specific word in a file.

Section D – Error & Exception Handling

24. Handle division by zero using try-except.

25. Handle invalid input using ValueError.

26. Demonstrate use of finally block.

27. Handle multiple exceptions in one program.

28. Create a custom exception for negative age input.

Section E – NumPy Practical

29. Create a NumPy array of 10 elements and print its shape.

30. Perform element-wise addition of two arrays.

31. Find mean, median, and standard deviation of an array.

32. Reshape a 1D array into 2D array.

33. Perform matrix multiplication using NumPy.

34. Demonstrate slicing and indexing in NumPy array.

35. Generate random numbers using NumPy.

Section F – Pandas Practical

36. Create a Pandas Series from a list.

37. Create a DataFrame from dictionary data.

38. Read a CSV file using Pandas.

39. Display first 5 and last 5 rows of dataset.

40. Filter rows based on a condition.

41. Handle missing values in a dataset.

42. Perform groupby operation on a dataset.

43. Merge two DataFrames.

44. Sort a DataFrame based on a column.

45. Export a DataFrame to CSV file.

Section G – Mini Data Analysis Task

46. Load a CSV dataset and display basic information.
47. Calculate descriptive statistics of dataset.
48. Identify and fill missing values.
49. Find top 5 records based on a specific column.
50. Perform a simple data analysis workflow and summarize insights.